Calculus Skeleton Notes

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Calculus Skeleton Notes

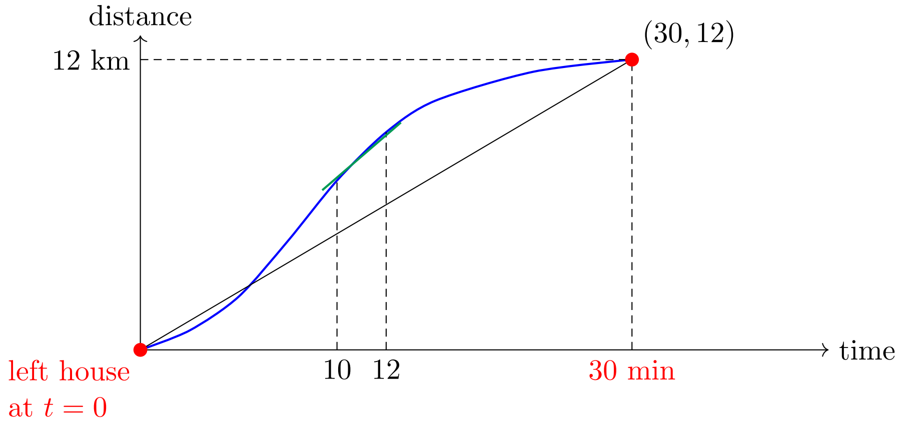
Goals: Compute average rates of change, slopes of secant lines, and motivate the idea of limits
Before Class Videos: None today, starting next class

Example 1: I left my house at time $t = 0$. 30 minutes later, I was at UVic which is 12 km away.

a) What was my average speed?

$$\text{Average Speed} = \frac{\Delta \text{ distance}}{\Delta \text{ time}} = \frac{12 \text{ km}}{0.5 \text{ h}} = 24 \text{ km/h}.$$

b) Sketch a possible graph



c) How could I approximate my speed at $t = 10$ minutes?
Starting at 10 min, I go 1 km in 2 min.

$$\text{Average speed} = \frac{1 \text{ km}}{2/60 \text{ h}} = 30 \text{ km/h}.$$

How can I approximate my speed at 10 min?

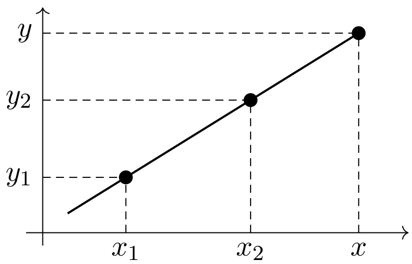
Compute average velocity over $[10, 11]$, $[10, 10.1]$, $[10, 10.01]$, etc

Tell LIDAR story

Future: "Need instantaneous velocity!"

Example 2: What is average rate of change of $f(x) = x^2$ on $[2, 4]$?
Average rate of change $\frac{f(4)-f(2)}{4-2} = \frac{16-4}{4-2} = 6$

Example 3: Find the equation of the line through (x_1, y_1) and (x_2, y_2).

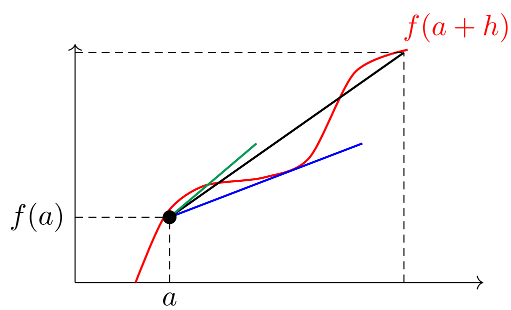


$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = m, \qquad m = \frac{y - y_1}{x - x_1}.$$

$$y - y_1 = m(x - x_1) \quad \Longrightarrow \quad y = y_1 + m(x - x_1) = mx + \underbrace{[y_1 - mx_1]}_{b}.$$

Example 4: Consider the graph of a function $f(x)$. How do I write an equation for the secant line through $(x_1, f(x_1))$ and $(x_2, f(x_2))$?

$$\text{Answer} \qquad y = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$$



Slope of secant line

$$= \frac{f(a+h) - f(a)}{(a+h) - a} = \frac{\text{rise}}{\text{run}}.$$

Big Idea: As x_2 gets closer to x_1 the slope of the secant line approaches the slope of the tangent line at x_1. We need to investigate and define this limiting process more precisely.

Suggested after class problems from APEX: None today, but review the Course Outline on Brightspace

Calculus Skeleton Notes

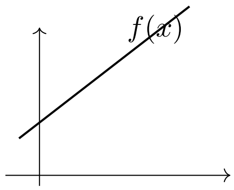
Goals: Describe limits graphically, numerically, and algebraically.

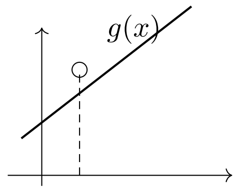
Before Class Videos: 1.1.1 and 1.1.2

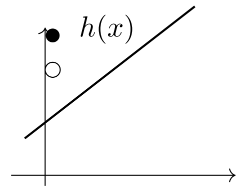
Example 1: Consider these three functions:

$$f(x) = x + 1, \qquad g(x) = \frac{x^2 - 1}{x - 1} = \frac{(x+1)(x-1)}{x-1}, \qquad h(x) = \begin{cases} x+1, & x \neq 1, \\ 3, & x = 1. \end{cases}$$

How are these similar or different?







Domain: $\qquad \mathbb{R}, \qquad (-\infty, 1) \cup (1, \infty), \qquad \mathbb{R}.$

Same everywhere but $x = 1$, where $f(1) = 2$, $g(1)$ undefined, $h(1) = 3$.

Definition:

$\lim_{x \to a} f(x) = L$ if $f(x)$ can be made arbitrarily close to L for all values of x sufficiently close to a.

Example 2: Compute the limit as $x \to 2$ of $f(x)$, $g(x)$, and $h(x)$.

$$\lim_{x \to 1} f(x) = \lim_{x \to 1} g(x) = \lim_{x \to 1} h(x) = 2$$

Example 3: Use a table of values to estimate $\lim_{x \to 1} (x + 1)$.

x	$f(x)$
0.9	1.9
0.99	1.99
0.999	1.999
1.1	2.1
1.01	2.01
1.001	2.001

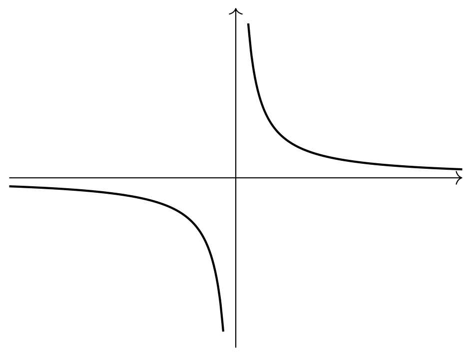
slope? closer to 2.

Example 4: Use a table of values to estimate $\lim_{x\to 0} \frac{1}{x}$.

x	$f(x)$
0.1	10
0.01	100
0.001	1000

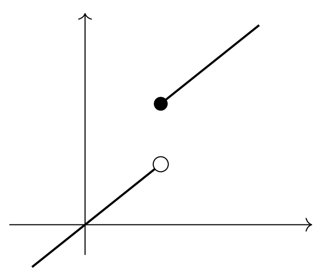
not getting close to anything!

So limit does not exist!



Example 5: Investigate

$$f(x) = \begin{cases} x, & x < 1, \\ x+1, & x \geq 1. \end{cases}$$



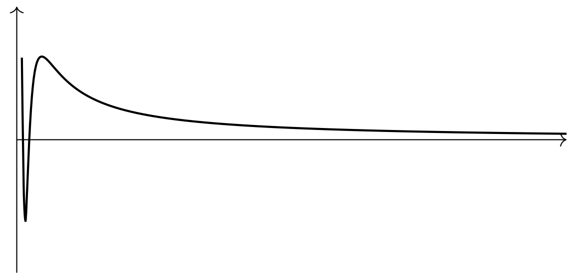
Definition:

$\lim_{x\to a^+} f(x)$ means $f(x)$ gets arbitrarily close to L for all x sufficiently close to a from the right, i.e. $x > a$.

$$\lim_{x\to 1^+} f(x) = \qquad \lim_{x\to 1^-} f(x) =$$

Example 6: Investigate limits of

$$f(x) = \sin\left(\frac{1}{x}\right)$$



$$\lim_{x \to 0^+} f(x) \text{ DNE}, \quad \text{as } f(x) \text{ gets close to every } y \text{ value in } [-1, 1].$$

Example 7: Graphically investigate limits of $\frac{x^2-1}{x^2-x-2}$

Suggested after class problems from APEX: Section 1.1 problems 9,11,15